

Normal (Gaussian) Distribution

[< Previous](#)[Next >](#)

Normal Distribution

The Normal Distribution is one of the most important distributions.

It is also called the Gaussian Distribution after the German mathematician Carl Friedrich Gauss.

It fits the probability distribution of many events, eg. IQ Scores, Heartbeat etc.

Use the `random.normal()` method to get a Normal Data Distribution.

It has three parameters:

`loc` - (Mean) where the peak of the bell exists.

`scale` - (Standard Deviation) how flat the graph distribution should be.

`size` - The shape of the returned array.

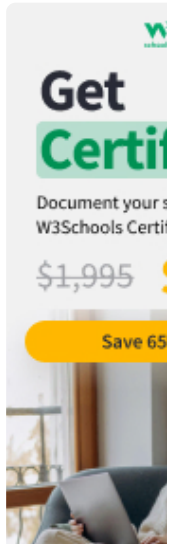
Example

[Get your own Python Server](#)

Generate a random normal distribution of size 2x3:

```
from numpy import random  
  
x = random.normal(size=(2, 3))  
  
print(x)
```

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Tutorials ▼

References ▼

Exercises ▼

Certificates ▼



⋮ In

HTML CSS JAVASCRIPT SQL PYTHON JAVA PHP HOW TO W3.CSS C C# >

NumPy Data Types

NumPy Copy vs View

NumPy Array Shape

NumPy Array Reshape

NumPy Array Iterating

NumPy Array Join

NumPy Array Split

NumPy Array Search

NumPy Array Sort

NumPy Array Filter

NumPy Random

Random Intro

Data Distribution

Random Permutation

Seaborn Module

Normal Distribution

Binomial Distribution

Poisson Distribution

Uniform Distribution

Logistic Distribution

Multinomial Distribution

Exponential Distribution

Chi Square Distribution

Rayleigh Distribution

Pareto Distribution

```
from numpy import random  
  
x = random.normal(loc=1, scale=2, size=  
(2, 3))  
  
print(x)
```

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Visualization of Normal Distribution

Example

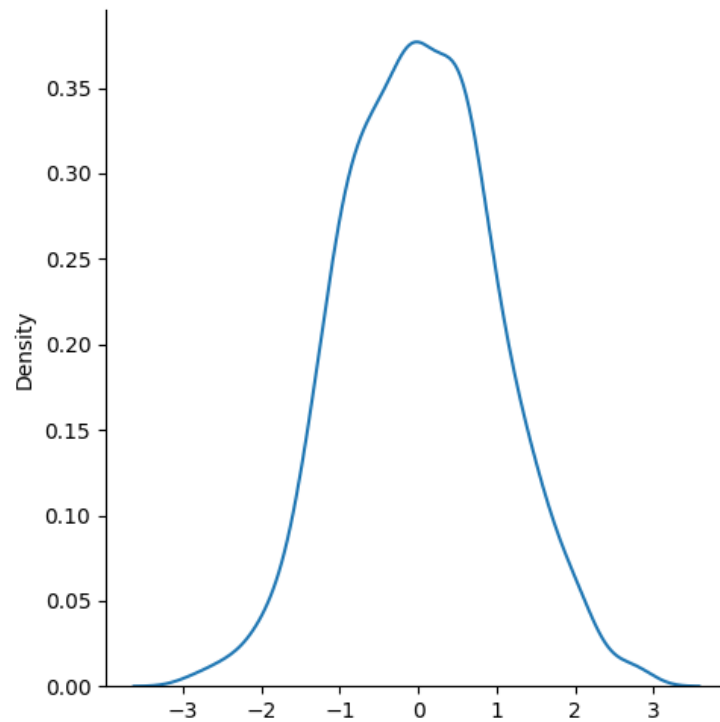
```
from numpy import random  
import matplotlib.pyplot as plt  
import seaborn as sns  
  
sns.displot(random.normal(size=1000),  
kind="kde")
```

[Zipf Distribution](#)`plt.show()`

NumPy ufunc

[ufunc Intro](#)[ufunc Create Function](#)[ufunc Simple Arithmetic](#)[ufunc Rounding Decimals](#)

Result

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Note: The curve of a Normal Distribution is also known as the Bell Curve because of the bell-shaped curve.

Exercise ?

The `random.normal()` method has three parameters, which ones?

- ☐ dept scale size
- ☐ mean dev size